

Optimization: PL condition

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1 Motivation

Example 1 (Quadratic programming). *The objective function is*

$$\min_{x \in \mathbb{R}^d} x^T D x, \text{ s.t. } x \geq 0. \quad (1)$$

We have following observations:

1. *Eq. (1) is a NP-hard problem.*
2. *D could have both positive and negative eigenvalues,*
3. *The gradient at $x = 0$ is zero. Therefore, $x = 0$ is a stationary point but not necessary a local minimizer.*

In fact, computationally verifying a point is a local minimizer is hard. Most optimization algorithms are looking for a stationary point.

2 Convex optimization

Definition 1 (Convex function). *A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex if for any x, y and $\lambda \in [0, 1]$, we have*

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

If f is differentiable, an equivalent definition is that for any x, y ,

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle.$$

If f is twice-differentiable, and $H(f)$ is the Hessian matrix of f , then another equivalent definition is $H(f) \succeq 0$.

Why convex programming is popular?

1. **Stationary point = Local minimizer = Global minimizer.**
2. Gradient descent algorithm guarantees linear convergence rate under mild conditions.

However, many problems are not convex, or even locally convex!

Example 2 (Non-linear least squares). Given n observations x_i, y_i , and a parametric function family $f(x; \theta)$, we hope to find the best fitting of the data. With a square loss function, the objective function is

$$\min_{\theta \in \mathbb{R}^m} \frac{1}{2} \sum_{i=1}^n (f(x_i; \theta) - y_i)^2. \quad (2)$$

When $n > m$, which is also known as the over-determined system, we know Eq. (2) is locally convex and classical convergence analysis applies. When $n < m$, i.e., the under-determined system, the landscape of Eq. (2) is nowhere local convex in general, as illustrated in Figure 1 (Liu et al., 2020).

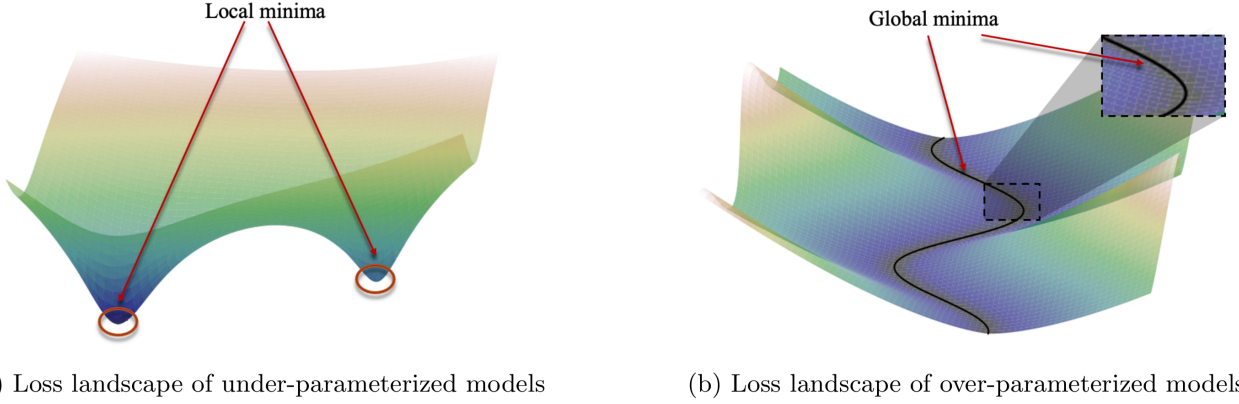


Figure 1: Panel (a): Loss landscape is locally convex at local minima. Panel (b): Loss landscape incompatible with local convexity as the set of global minima is not locally linear.

3 PL condition

Any milder assumption that is satisfied in a broader range of problems, and can have fast convergence rate? Yes!

Definition 2 (Polyak-Łojasiewicz Condition). A function f satisfies PL condition if there exists some $\mu > 0$ such that for any x , we have

$$\frac{1}{2\mu} \|\nabla f(x)\|_2^2 \geq f(x) - f(x^*),$$

where $x^* = \arg \min_x f(x)$.

We can let $f(x) = f(x) - f(x^*)$ to eliminate the last term in the definition. It is not hard to find that PL condition inherits some nice properties of convexity condition.

1. Stationary point = Local minimizer = Global minimizer.
2. Linear convergence rate using GD/SGD (Karimi et al., 2016).
3. Convexity implies PL condition.

Let us revisit Example 2. Let $f(\theta) = (f(x_1; \theta) - y_1, \dots, f(x_n; \theta) - y_n)^\top : \mathbb{R}^m \rightarrow \mathbb{R}^n$. If this non-linear least squares loss satisfies PL condition, it means

$$\begin{aligned} \frac{1}{2\mu} \|\nabla(L \circ f)(\theta)\|_2^2 \geq L \circ f(\theta) &\iff \frac{1}{2\mu} \|f(\theta)^\top J_f(\theta)\|_2^2 \geq \|f(\theta)\|_2^2 \\ &\iff \min_{\theta} \lambda_{\min}(J_f(\theta)J_f(\theta)^\top) \geq 2\mu. \end{aligned}$$

Here, $J_f = (\partial f_i(\theta)/\partial \theta_j) \in \mathbb{R}^{n \times m}$ is the Jacobian matrix of f w.r.t. θ , $\lambda_{\min}$ means the smallest eigenvalue.

Therefore, as long as $J_f(\theta)J_f(\theta)^\top$ has a lower bound, PL condition will hold. In practice, we do not need it hold for the entire parameter space, a neighborhood of initialization of θ would be enough for GD.

References

- Hamed Karimi, Julie Nutini, and Mark Schmidt. Linear convergence of gradient and proximal-gradient methods under the polyak-łojasiewicz condition. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pages 795–811. Springer, 2016.
- Chaoyue Liu, Libin Zhu, and Mikhail Belkin. Loss landscapes and optimization in over-parameterized non-linear systems and neural networks. *arXiv preprint arXiv:2003.00307*, 2020.